

Differential Gene Expression Analysis of TCGA Lung Squamous Cell Carcinoma (LUSC) RNA-seq Data Using a Python-Based Bioinformatics Pipeline

Abstract

Lung cancer is one of the leading causes of cancer-related mortality worldwide, with Lung Squamous Cell Carcinoma (LUSC) representing a major subtype of non-small cell lung cancer. The aim of this project was to investigate transcriptomic differences between tumour and normal lung tissue using RNA-sequencing (RNA-seq) data from The Cancer Genome Atlas (TCGA).

Gene expression data were processed using a Python-based bioinformatics workflow including preprocessing, normalization, exploratory analysis, and differential gene expression testing. Principal Component Analysis (PCA) was applied to evaluate global transcriptomic structure and sample clustering. Differential expression analysis identified genes significantly dysregulated between tumour and normal tissues using statistical testing with false discovery rate (FDR) correction.

The analysis demonstrated clear separation between tumour and normal samples and revealed extensive transcriptional reprogramming in LUSC. A large number of genes were significantly upregulated or downregulated, consistent with molecular alterations associated with oncogenesis. This project illustrates how computational methods can be applied to publicly available cancer genomics data to extract biologically meaningful insights.

1. Introduction

RNA-sequencing (RNA-seq) is a high-throughput sequencing technology that allows quantification of gene expression across the entire transcriptome. It has become a standard method for studying cancer biology, biomarker discovery, and molecular classification of tumors.

Lung Squamous Cell Carcinoma (LUSC) is a subtype of non-small cell lung cancer that arises from epithelial cells of the lung airways. Molecular characterization of LUSC can provide insight into mechanisms of tumor development and potential therapeutic targets.

The Cancer Genome Atlas (TCGA) provides publicly available genomic datasets for multiple cancer types, including RNA-seq expression profiles. By comparing tumor and normal tissue gene expression, it is possible to identify genes associated with tumorigenesis.

The aim of this project was to:

- evaluate transcriptomic differences between tumor and normal samples
- visualize gene expression patterns
- identify differentially expressed genes (DEGs)

2. Materials and Methods

2.1 Dataset

RNA-seq gene expression data were obtained from The Cancer Genome Atlas (TCGA) Lung Squamous Cell Carcinoma cohort. The dataset contained expression measurements for approximately 20,000 protein-coding genes across tumour and normal lung tissue samples. The raw sequencing files were not used directly. Instead, a processed expression matrix was analysed.

2.2 Data Preprocessing

The analysis was performed using Python. The processed expression matrix was imported, and sample labels (tumour vs normal) were assigned. Low-expression genes were filtered to reduce noise and improve statistical power. Expression values were log2-transformed to stabilize variance across genes.

Normalization was necessary because RNA-seq expression values span several orders of magnitude and are influenced by sequencing depth. Proper preprocessing ensures that observed differences reflect biological variation rather than technical effects.

2.3 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was used as an unsupervised dimensionality reduction technique to explore global gene-expression structure. PCA transforms high-dimensional gene expression data into orthogonal components that capture the largest sources of variance.

PCA was used to:

- assess sample clustering
- detect potential outliers
- evaluate separation between biological conditions

The first two principal components were visualized to examine transcriptomic differences between tumour and normal samples.

2.4 Differential Gene Expression Analysis

Differential expression analysis was performed to identify genes whose expression significantly differed between tumour and normal samples.

Statistical testing was applied for each gene, and multiple testing correction was performed using False Discovery Rate (FDR).

Significance thresholds:

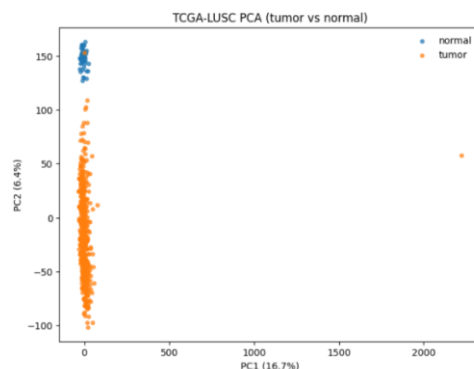
- Adjusted p-value < 0.05
- $|\log_2 \text{fold change}| > 1$

Genes passing both criteria were considered significantly differentially expressed.

2.5 Visualization

The following visualizations were generated:

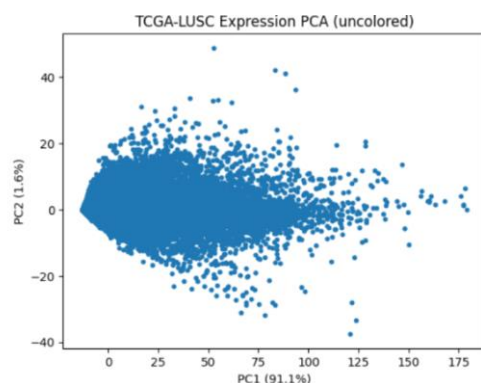
PCA scatter plot (Principal Component Analysis) of TCGA-LUSC gene-expression data showing tumour vs normal sample clustering



This PCA plot shows the separation of tumour and normal samples in the TCGA-LUSC dataset based on global gene-expression patterns. Each point represents one sample, with blue indicating normal tissue and orange indicating tumour tissue. PC1 explains 16.7% of the variance, while PC2 explains 6.4%, suggesting that variability is now distributed across multiple components. Unlike the earlier uncoloured PCA, the samples now form distinct groups. Normal samples cluster tightly together, indicating relatively consistent gene expression among healthy lung tissues. Tumour samples are more widely dispersed, reflecting biological heterogeneity within lung squamous cell carcinoma. The clear separation between most tumour and normal samples indicates

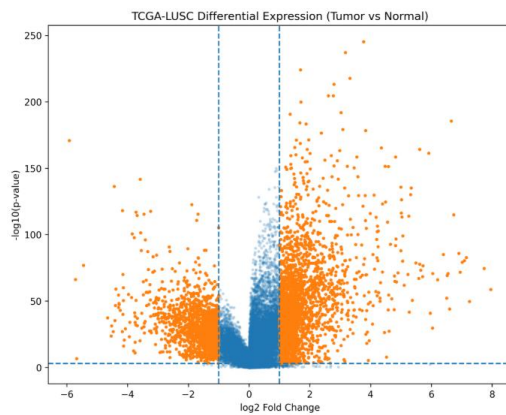
strong transcriptional differences between the two conditions. This supports the validity of the differential expression analysis results. One tumour sample appears as an extreme outlier far along PC1, which may represent a technical artifact or unusually distinct tumour biology. Such outliers should be checked because they can influence downstream statistical analyses. Overall, the PCA demonstrates successful normalization and reveals biologically meaningful structure in the dataset.

PCA projection plot



This PCA plot shows the distribution of TCGA-LUSC samples based on their global gene-expression profiles after dimensionality reduction. Each point represents one patient tumour sample, and the axes (PC1 and PC2) summarize the major patterns of variation across thousands of genes. PC1 explains 91.1% of the total variance, indicating that one dominant factor strongly influences the dataset. Because such a high proportion is captured by a single component, the variation is likely driven by a technical effect rather than biological subtype differences. The wedge- or comet-shaped spread of samples along PC1 suggests that sequencing depth or library size differences are influencing the expression values. This pattern is typical when PCA is performed on raw or insufficiently normalized RNA-seq counts. PC2 explains only 1.6% of the variance, meaning little additional structured variation is visible after the primary effect. As a result, potential biological groupings (such as tumour heterogeneity or molecular subtypes) are likely masked. Several samples appear as outliers on the far right of PC1, which may correspond to unusually high read counts or technical artifacts. Therefore, the PCA currently reflects technical variability more than biological signal. Applying a variance-stabilizing normalization (e.g., DESeq2 VST or rlog) should reduce this effect and reveal meaningful clustering among tumor samples.

Volcano plot



This volcano plot displays differential gene expression between tumour and normal samples in the TCGA-LUSC dataset. Each point represents a gene, positioned by its log₂ fold change on the x-axis and statistical significance ($-\log_{10}$ p-value) on the y-axis. Genes on the right side are upregulated in tumours, while genes on the left side are downregulated in tumours relative to normal tissue. The dashed vertical lines indicate the fold-change threshold, and the horizontal dashed line represents the p-value significance cutoff. Orange points correspond to genes that pass both the fold-change and significance criteria, identifying them as differentially expressed genes (DEGs). A large number of significant genes are observed, suggesting substantial transcriptional differences between tumour and normal lung tissue. There is a noticeable asymmetry with more strongly upregulated genes than downregulated ones. Highly significant genes appear at the top of the plot, indicating very small p-values and strong statistical confidence. Genes near the centre with low fold change and low significance likely represent background expression variation. Some genes show extremely large fold changes, which may correspond to tumour-specific markers or pathway activation. Overall, the results indicate clear molecular differences between cancerous and normal samples and support the validity of the differential expression analysis.

3. Results

3.1 Exploratory Data Analysis

PCA revealed clustering of samples according to biological condition. Normal samples formed a compact cluster, while tumour samples displayed greater dispersion, reflecting tumour heterogeneity. The separation between groups indicates substantial global transcriptomic differences between cancerous and healthy tissue.

3.2 Differential Expression

Differential expression analysis identified numerous significantly dysregulated genes between tumour and normal samples. The volcano plot showed genes with large fold changes and strong statistical significance. Both upregulated and downregulated genes were observed, indicating widespread transcriptional alterations in LUSC.

3.3 Biological Interpretation

The observed gene expression changes are consistent with known hallmarks of cancer, including:

- increased cellular proliferation
- altered metabolism
- dysregulation of cell cycle pathways
- changes in transcriptional regulation

Such widespread dysregulation reflects oncogenic transformation and tumour progression.

4. Discussion

The results demonstrate that transcriptomic profiling using RNA-seq data can clearly distinguish tumour from normal lung tissue. The observed separation in PCA indicates that disease status is a major contributor to gene expression variance. The large number of differentially expressed genes suggests widespread disruption of regulatory pathways in LUSC.

The large number of differentially expressed genes suggests multiple pathways are altered in LUSC. These genes may serve as potential biomarkers or therapeutic targets, although functional validation would be required.

Limitations:

- analysis performed on processed expression data
- no pathway enrichment analysis performed
- no independent validation dataset used

Despite these limitations, the workflow demonstrates a typical bioinformatics RNA-seq analysis pipeline.

5. Conclusion

This study successfully analysed TCGA LUSC RNA-seq gene expression data using a computational workflow. Tumour and normal samples showed clear transcriptomic separation, and numerous significantly dysregulated genes were identified.

The project demonstrates how bioinformatics methods can be applied to cancer genomics data to generate biologically meaningful insights.

6. Tools and Technologies

All analyses were performed in a Jupyter Notebook environment.

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Jupyter Notebook

7. Future Work

Future analysis could include:

- Gene Ontology enrichment analysis
- pathway analysis (KEGG)
- machine learning classification models
- biomarker discovery

8. References

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